OIL CONTROL VALVE (OCV) INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

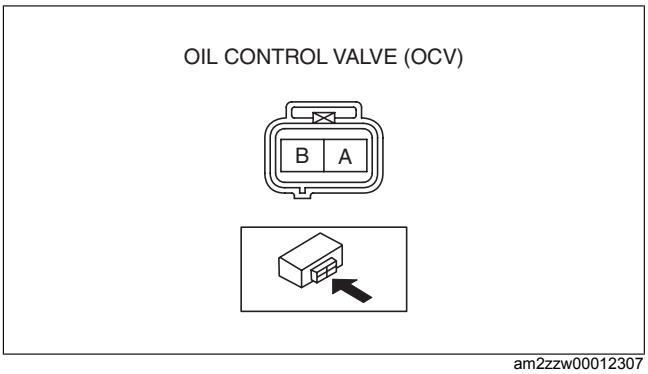
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SKYACTIV-G 1.3, SKYACTIV-G 1.5 (With 4-1 Exhaust System)
Coil Resistance Inspection

1. Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/ CONNECTION.)
2. Remove the plug hole plate. (See PLUG HOLE PLATE REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
3. Disconnect the OCV connector. (See OIL CONTROL VALVE (OCV) REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
4. Measure the resistance between terminals A and B using an ohmmeter.

   OCV coil resistance
   6.9—7.5 ohms [20 °C {68 °F}]

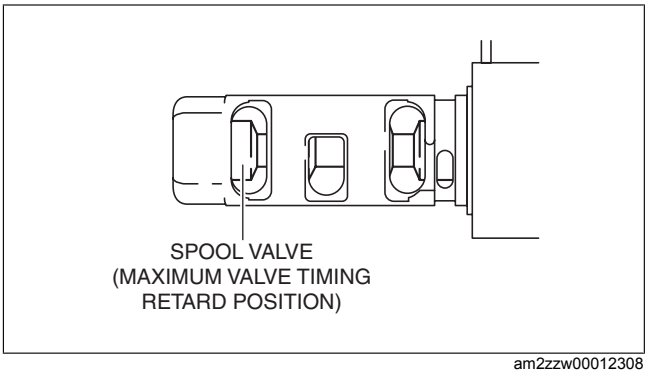
   - If not as specified, replace the OCV. (See OIL CONTROL VALVE (OCV) REMOVAL/ INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
5. Install in the reverse order of removal.



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Spool Valve Operation Inspection

1. Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/ CONNECTION.)
2. Remove the plug hole plate. (See PLUG HOLE PLATE REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
3. Remove the ignition coil. (See IGNITION COIL/ION SENSOR REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
4. Remove the cylinder head cover. (See TIMING CHAIN REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
5. Remove the OCV. (See OIL CONTROL VALVE (OCV) REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
6. Verify that the spool valve in the OCV is in the maximum valve timing retard position as indicated in the figure.
   - If not as specified, replace the OCV. (See OIL CONTROL VALVE (OCV) REMOVAL/ INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
7. Verify that the battery is fully charged. (See BATTERY INSPECTION.)
   - If not as specified, recharge the battery. (See BATTERY RECHARGING.)

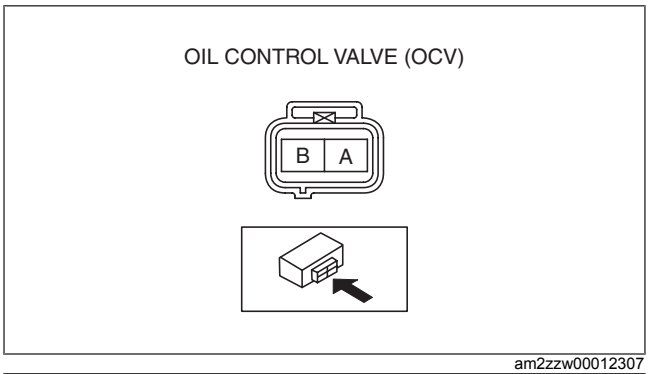


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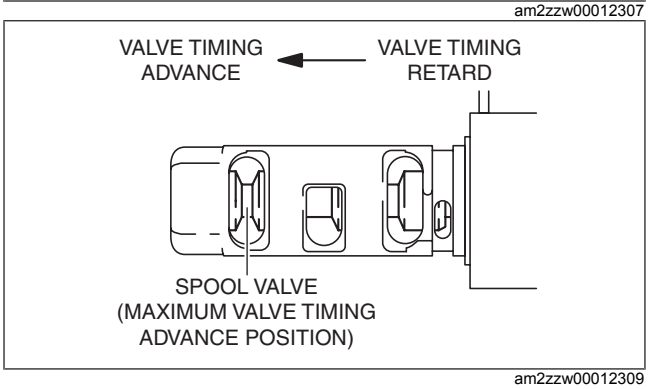
Note

- When applying battery positive voltage between the OCV terminals, the connection can be either of the following:
  — Positive battery cable to terminal A, negative battery cable to terminal B
  — Positive battery cable to terminal B, negative battery cable to terminal A

8. Apply battery positive voltage between the OCV terminals and verify that the spool valve operates and moves to the maximum valve timing advance position.
   - If not as specified, replace the OCV. (See OIL CONTROL VALVE (OCV) REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
9. Stop applying battery positive voltage and verify that the spool valve returns to the maximum valve timing retard position.
   - If not as specified, replace the OCV. (See OIL CONTROL VALVE (OCV) REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
10. Install the OCV. (See OIL CONTROL VALVE (OCV) REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)



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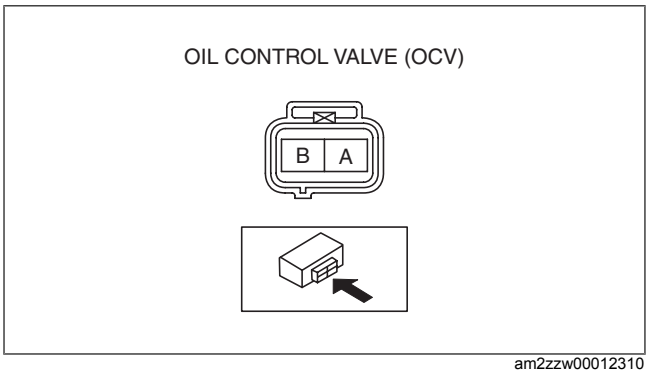
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SKYACTIV-G 1.5 (With 4-2-1 Exhaust System)
Coil Resistance Inspection

1. Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.)
2. Remove the plug hole plate. (See PLUG HOLE PLATE REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
3. Disconnect the OCV connector. (See OIL CONTROL VALVE (OCV) REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
4. Measure the resistance between terminals A and B using an ohmmeter.

   OCV coil resistance
   6.9—7.5 ohms [20 °C {68 °F}]

   - If not as specified, replace the OCV. (See OIL CONTROL VALVE (OCV) REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
5. Install in the reverse order of removal.

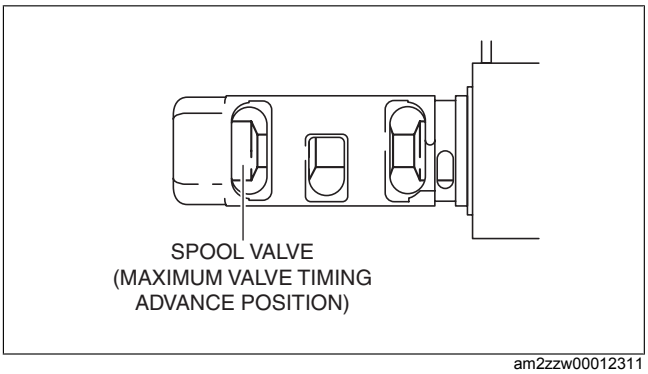


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Spool Valve Operation Inspection

1. Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.)
2. Remove the plug hole plate. (See PLUG HOLE PLATE REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
3. Remove the ignition coil/ion sensors. (See IGNITION COIL/ION SENSOR REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
4. Remove the cylinder head cover. (See TIMING CHAIN REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
5. Remove the OCV. (See OIL CONTROL VALVE (OCV) REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)

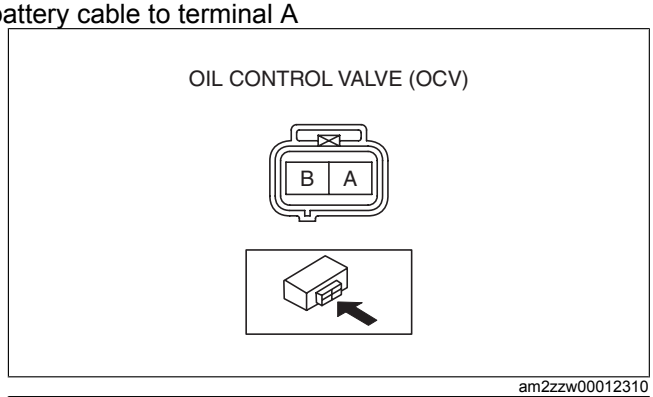
6. Verify that the spool valve in the OCV is in the maximum valve timing advance position as indicated in the figure.
   - If not as specified, replace the OCV. (See OIL CONTROL VALVE (OCV) REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
7. Verify that the battery is fully charged. (See BATTERY INSPECTION.)
   - If not as specified, recharge the battery. (See BATTERY RECHARGING.)



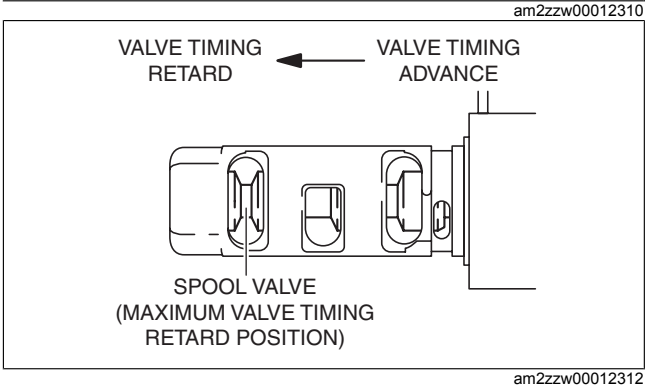
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Note
- When applying battery positive voltage between the OCV terminals, the connection can be either of the following:
  — Positive battery cable to terminal A, negative battery cable to terminal B
  — Positive battery cable to terminal B, negative battery cable to terminal A

8. Apply battery positive voltage between the OCV terminals and verify that the spool valve operates and moves to the maximum valve timing retard position.
   - If not as specified, replace the OCV. (See OIL CONTROL VALVE (OCV) REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
9. Stop applying battery positive voltage and verify that the spool valve returns to the maximum valve timing advance position.
   - If not as specified, replace the OCV. (See OIL CONTROL VALVE (OCV) REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
10. Install the OCV. (See OIL CONTROL VALVE (OCV) REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)



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